

exp 3.py

```
1 import pandas as pd
2 from sklearn.preprocessing import LabelEncoder
3 from sklearn.tree import DecisionTreeClassifier
4 data = pd.DataFrame({
5     "Outlook": ["Sunny", "Sunny", "Overcast", "Rain", "Rain", "Rain", "Overcast", "Sunny", "Sunny", "Rain", "Sunny", "Overcast", "Overcast", "Rain"],
6     "Temperature": ["Hot", "Hot", "Hot", "Mild", "Cool", "Cool", "Cool", "Mild", "Cool", "Mild", "Mild", "Mild", "Hot", "Mild"],
7     "Humidity": ["High", "High", "High", "High", "Normal", "Normal", "Normal", "High", "Normal", "Normal", "Normal", "Normal", "High", "Normal", "High"],
8     "Wind": ["Weak", "Strong", "Weak", "Weak", "Weak", "Strong", "Strong", "Weak", "Weak", "Weak", "Strong", "Strong", "Weak", "Strong"],
9     "PlayTennis": ["No", "No", "Yes", "Yes", "Yes", "No", "Yes", "No", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "No"]
10 })
11 X = data.iloc[:, :-1]
12 y = data.iloc[:, -1]
13 encoders = {}
14 for column in X.columns:
15     le = LabelEncoder()
16     X[column] = le.fit_transform(X[column])
17     encoders[column] = le
18 target_encoder = LabelEncoder()
19 y = target_encoder.fit_transform(y)
20 model = DecisionTreeClassifier(criterion="entropy")
21 model.fit(X, y)
22 print("Decision Tree trained successfully.")
23 sample = pd.DataFrame({
24     "Outlook": ["Sunny"], "Temperature": ["Cool"], "Humidity": ["Normal"], "Wind": ["Weak"]
25 })
26 for column in sample.columns:
27     sample[column] = encoders[column].transform(sample[column])
28 prediction = target_encoder.inverse_transform(model.predict(sample))
29 print("Prediction:", prediction[0])
```

Run exp 3



```
↑ /Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 3.py
Decision Tree trained successfully.
> Prediction: Yes
```